RIVAROXABAN

Take 15/20 mg with food

Not for mechanical valves

CHEMICAL NAME 5-chloro-N-({(5S)-2-oxo-3-[4-(3-oxomorpholin-4-yl)phenyl]-1,3-oxazolidin-5-yl}methyl)thiophene-2-carboxamide

CLASS Direct oral anticoagulant (DOAC) — direct factor Xa inhibitor

TRADE NAMES Xarelto · Rivarox · Rivaxa

THERAPEUTIC USES / INDICATIONS

- Stroke prevention in non-valvular atrial fibrillation.
- Treatment of DVT/PE and prevention of recurrence.
- VTE prophylaxis after hip/knee replacement.
- With aspirin: CAD/PAD (2.5 mg twice daily).

ADMINISTRATION

- **PO**; 15 mg and 20 mg doses must be taken with food (↑ absorption); 10 mg with or without food.
- Can be crushed and given in water/apple purée or via NG tube.
- No routine INR monitoring.

DOSAGES

- **AF:** 20 mg once daily with evening meal (15 mg if CrCl 15–49).
- **DVT/PE:** 15 mg twice daily for 21 days, then 20 mg once daily.
- **Extended prevention:** 10 mg (or 20 mg) once daily after 6 months.
- **Hip/knee replacement:** 10 mg once daily (5 weeks hip, 2 weeks knee).
- Avoid if CrCl <15.

PHARMACOKINETICS

- **Onset:** 2–4 h (peak). **Duration:** ~24 h.
- Bioavailability ~66% fasting (~100% with food for 20 mg); metabolised by CYP3A4 and P-gp.
- Excreted in urine (1/3 unchanged) and faeces. **t½:** 5–9 h (11–13 h elderly).

PHARMACODYNAMICS

- Directly and reversibly inhibits **free and clot-bound factor Xa** → ↓ thrombin generation.
- No antithrombin needed; prolongs PT (not reliable for monitoring).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Bleeding: gums, nose, bruising, menorrhagia.
- Nausea, dyspepsia.
- Anaemia.

ADVERSE EFFECTS (SERIOUS)

- Major bleeding (GI, intracranial).
- Spinal/epidural haematoma.
- Liver injury (rare).
- Reversal: andexanet alfa or prothrombin complex concentrate.

CONTRAINDICATIONS & CAUTIONS

- Active clinically significant bleeding.
- Mechanical heart valves; moderate–severe rheumatic mitral stenosis.
- Hepatic disease with coagulopathy (Child-Pugh B/C).
- Pregnancy, breastfeeding; antiphospholipid syndrome (triple positive).

KEY INTERACTIONS

- **Strong CYP3A4 + P-gp inhibitors** (ketoconazole, itraconazole, ritonavir): avoid.
- **Strong inducers** (rifampicin, carbamazepine, phenytoin, St John's wort): avoid; ↓ effect.
- **Antiplatelets, NSAIDs, SSRIs:** ↑ bleeding.

NURSING MANAGEMENT

- Check renal function (CrCl), LFTs, CBC at baseline and periodically.
- Give 15/20 mg with food.
- Monitor for bleeding.
- Peri-procedure: follow stop/restart plan (usually 24–48 h before).
- Neuraxial anaesthesia timing rules.

PATIENT EDUCATION

- Take with your main meal (15 mg and 20 mg).
- Do not miss doses — protection fades within a day.
- Missed dose: take same day if remembered; do not double (once-daily regimen).
- Report bleeding; carry alert card; tell dentists/surgeons.

KEY POINTS

- Direct Xa inhibitor (DOAC).
- Not for mechanical valves or pregnancy.
- AF 20 mg OD with food; VTE 15 mg BD × 21 days.
- Reverse: andexanet alfa / PCC.

SOURCES CHECKED **[421]** Rivaroxaban 10 mg Film coated tablet (UK SmPC) **[422]** Xarelto, oral (US label) **[423]** Xarelto, oral (US label, Janssen Pharmaceuticals)

Full references p. 432–457